Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

Yonghan Jung[1*] **Min Woo Park**[2] **Sanghack Lee**[2*]
[1]Purdue University [2]Seoul National University
jung222@purdue.edu {alsdn0110,sanghack}@snu.ac.kr

Abstract

Covariate adjustment is a fundamental tool in causal inference. This corrected version repairs the sequential adjustment criterion (SAC) proposed in the NeurIPS 2024 paper. The published forbidden set is too restrictive: its directed paths and descendants must be computed after intervening on later treatments, as the published separation condition already requires. With this correction, SAC is sound and complete for the admissible sequential adjustment formulas in a known causal DAG or acyclic directed mixed graph. Sufficiency follows from conditional adjustment. For necessity, we isolate later treatments and lift an ordinary-adjustment Gaussian countermodel to the final sequential formula; averaging the earlier history preserves its incorrect treatment slope. Thus every failed criterion has a positive linear Gaussian countermodel, although validity is nonparametric. We also repair the canonical construction and replace simultaneous pruning with a forward algorithm using finalized histories and required conditioning sets. It finds a valid sequence whenever one exists and returns an inclusion-minimal sequence in $O(m(|\mathbf{V}|+|\mathbf{E}|))$ time. These results restore the completeness and construction guarantees for the stated formula class; they do not address variance optimality.

1 Introduction

Covariate adjustment (also known as the back-door adjustment (Pearl, 1995)) —$\sum_{\mathbf{z}} P(y \mid x, \mathbf{z})P(\mathbf{z})$ where $\mathbf{Z}$ is a set of covariates, X is a treatment, and Y is an outcome—is one of the most prevalent method for estimating causal effects from observational data. Sufficient conditions have been established to determine whether causal effects can be expressed through covariate adjustment. For example, back-door criterion (BD) (Pearl, 1995) is a sound graphical condition to determine if the causal effect is expressible as a covariate adjustment. Shpitser et al. (2010) extended the BD and developed a sound and complete graphical criterion, called adjustment criterion (AC). van der Zander et al. (2014) and Maathuis and Colombo (2015) extended the adjustment criterion to general graph classes including directed acyclic graphs, maximal ancestral graphs, or partial ancestral graphs. The criterion from Maathuis and Colombo (2015) is sound but incomplete, which Perkovic et al. (2018) addressed by developing a sound and complete adjustment criterion.

Beyond such static adjustment criterion, *sequential* covariate adjustment (also known as the *g-formula* (Robins, 1986)) has been studied. For example, *sequential back-door (SBD) criterion* (Pearl and Robins, 1995) extended the back-door criterion to the sequential setting. Since the SBD criterion only considered a single outcome, Jung et al. (2020) extended it to accommodate multiple outcome variables called *multi-outcome sequential back-door (mSBD)* criterion. However, as detailed in Sec. 3, this criterion is not complete; i.e., there exists an example where the mSBD criterion is not satisfied, but the causal effect is identified as a sequential covariate adjustment. Sequential ignorability provides

*Corresponding authors

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Table 1: Comparison of graphical criteria for (sequential) covariate adjustments

Criterion	Static	Sequential	Multi-outcome	Completeness
BD (Pearl, 1995, 2000)	✓	✗	N/A	✗
AC (Shpitser et al., 2010)	✓	✗	N/A	✓
SBD (Pearl and Robins, 1995)	✓	✓	✗	✗
mSBD (Jung et al., 2020)	✓	✓	✓	✗
SAC (corrected)	✓	✓	✓	✓

sufficient conditions for a g-formula representation, but the treatment-descendant covariates permitted here require a separate analysis; see Sec. 3.2.

The corrected sequential adjustment criterion is sound and complete for the admissible formula class defined below. Table 1 compares it with previous graphical criteria for covariate adjustment.[2] The corrected results are as follows:

1. We demonstrate the incompleteness of mSBD criterion by exemplifying a graph such that the causal effect is given as sequential covariate adjustment where the mSBD criterion is not satisfied. We also discuss representation and verification considerations for other criteria.
2. We correct SAC by computing its forbidden set in the graph intervened on later treatments. A replacement proof establishes soundness and completeness for the final, marginalized formula. Whenever SAC fails, a compatible positive linear Gaussian model witnesses the failure at the mean of one outcome coordinate.
3. We construct a valid admissible sequence whenever one exists. We repair simultaneous pruning by constructing each block from the finalized history and preserving required conditioning variables. The corrected procedure returns a stagewise and componentwise inclusion-minimal sequence in linear time per stage.

The published separation graph already intervenes on future treatments; the change to Definition 7 is the graph used for forbidden descendants. The four-node example in Sec. 4 refutes the published condition, while Lemma S.2 proves necessity of its replacement without assuming that earlier stages are valid. These corrections restore guarantees claimed in the original paper. Completeness concerns a fixed known causal graph, scalar treatments in a fixed topological order, the graph-derived outcome partition, and admissible covariate blocks. The universal model class includes real-valued Gaussian models; a separate completeness theorem for prescribed finite variable domains is not asserted.

The proofs for all results are provided in Sec. A in the supplementary document.

2 Preliminaries

We introduce the necessary notation and background. Key notations are summarized in Table 2.

Structural Causal Models. We use Structural Causal Models (SCMs) (Pearl, 2000). An SCM $\mathcal{M}$ is a quadruple $\mathcal{M} \coloneqq \langle \mathbf{U}, \mathbf{V}, P(\mathbf{U}), F \rangle$. $\mathbf{U}$ is a set of exogenous (latent) variables following a joint distribution $P(\mathbf{U})$. $\mathbf{V}$ is a set of endogenous (observable) variables whose values are determined by functions $F = \{f_{V_i}\}_{V_i \in \mathbf{V}}$ such that $V_i \leftarrow f_{V_i}(\mathbf{pa}_i, \mathbf{u}_i)$ where $\mathbf{PA}_i \subseteq \mathbf{V} \setminus \{V_i\}$ and $\mathbf{U}_i \subseteq \mathbf{U}$. Each SCM $\mathcal{M}$ induces a distribution $P(\mathbf{V})$ and a causal graph $\mathcal{G} = \mathcal{G}(\mathcal{M})$ over $\mathbf{V}$ in which there exists a directed edge from every variable in $\mathbf{PA}_i$ to V_i and dashed-bidirected arrows encode correlated latent variables. Performing an intervention fixing $\mathbf{X} = \mathbf{x}$ is represented through the do-operator, $\mathrm{do}(\mathbf{X} = \mathbf{x})$, which encodes the operation of replacing the original equations of X (i.e., $f_X(\mathbf{pa}_x, \mathbf{u}_x)$) by the constant x for all $X \in \mathbf{X}$ and induces an interventional distribution $P(\mathbf{V} \mid \mathrm{do}(\mathbf{x}))$.

Graphs. We consider a known causal DAG or an acyclic directed mixed graph (ADMG) $\mathcal{G}$ having vertices $\mathbf{V}$ and edges $\mathbf{E}$ composed of directed ($V_i \to V_j$) and bidirected edges ($V_i \leftrightarrow V_j$), including

[2]The multi-outcome column indicates whether the outcome for each criterion is a sequence of vectors or a single vector. Therefore, BD and AC, which deal with static context, are not applicable.

Table 2: Key notations

Notation	Definition
$\mathbf{X}, X, \mathbf{x}, x$	A random vector, variable, and their realized values.
$\sum_{\mathbf{x}} f(\mathbf{x})$	A marginalization of $f(\mathbf{x})$ over values in the domain of $\mathbf{X}$.
$\mathbf{X}^{(i)}$	$(\mathbf{X}_0, \cdots, \mathbf{X}_i)$ for the ordered set $\mathbf{X} = (\mathbf{X}_0, \cdots, \mathbf{X}_n)$.
$\mathbf{X}^{\geq i}$	$(\mathbf{X}_i, \cdots, \mathbf{X}_n)$ for the ordered set $\mathbf{X}$.
$\mathrm{De}_{\mathcal{G}}(\mathbf{C}), \mathrm{An}_{\mathcal{G}}(\mathbf{C})$	Descendants and ancestors of $\mathbf{C}$ in $\mathcal{G}$.
$\mathrm{pcp}_{\mathcal{G}}(\mathbf{X}, \mathbf{Y})$	Proper causal path set; i.e., $(\mathrm{De}_{\mathcal{G}_{\overline{\mathbf{X}}}}(\mathbf{X}) \setminus \mathbf{X}) \cap \mathrm{An}_{\mathcal{G}_{\underline{\mathbf{X}}}}(\mathbf{Y})$.
$\mathrm{dpcp}_{\mathcal{G}}(\mathbf{X}, \mathbf{Y})$	$\mathrm{De}_{\mathcal{G}}(\mathrm{pcp}_{\mathcal{G}}(\mathbf{X}, \mathbf{Y}))$.

bows but no directed cycles. An ADMG here is a causal latent-projection graph of an acyclic latent-variable model, not a MAG or PAG. Ancestors and descendants include the vertex itself. We use "d-separation" for the corresponding mixed-graph path criterion (also called m-separation). We assume a compatible acyclic causal model and positivity wherever an observational conditional in an adjustment formula is evaluated. Universal validity quantifies over compatible positive models with unrestricted variable domains, including positive linear Gaussian models. Sufficiency applies to discrete and continuous models; necessity does not separately cover a model class restricted to prescribed finite alphabets. Sums may be replaced by integrals, and continuous conditional equalities have their usual almost-everywhere interpretation. Replacing each bidirected edge by an independent latent parent of its endpoints supplies a compatible latent DAG for countermodels and graphical separation calculations. We do not assume that this particular expansion represents every compatible distribution, or that latent mechanisms must have positive densities. An ordered sequence of distinct nodes in $\mathcal{G}$ is called a *path* between V_i and V_j in $\mathcal{G}$ if (1) the start node is V_i and the end node is V_j, and (2) there is an edge (directed or bidirected) between any two subsequent variables in the sequence. A variable V_i is called a *collider* on the path if the path contains two edges having arrow heads toward V_i; i.e., $* \!\!\rightarrow V_i \leftarrow \!\!*$ where $* \!\!\rightarrow \in \{\rightarrow, \leftrightarrow\}$. A path is *directed* (or causal), if it contains only directed edges, all pointing in the same direction. We say that two sets of vertices $\mathbf{A}, \mathbf{B} \subseteq \mathbf{V}$ are *d-connected* w.r.t. $\mathbf{Z} \subseteq \mathbf{V} \setminus (\mathbf{A} \cup \mathbf{B})$ in $\mathcal{G}$ if there exists a path between $V_i \in \mathbf{A}$ and $V_j \in \mathbf{B}$ where every non-colliders on the path is not contained in $\mathbf{Z}$ and all colliders are in $\mathrm{An}_{\mathcal{G}}(\mathbf{Z})$. We say that $\mathbf{Z}$ *d-separates* $(\mathbf{A}, \mathbf{B})$ in $\mathcal{G}$ if they are not d-connected w.r.t. $\mathbf{Z}$ in $\mathcal{G}$. We denote by $\mathcal{G}_{\overline{\mathbf{A}}\underline{\mathbf{B}}}$ the graph obtained from $\mathcal{G}$ by removing all incoming arrowheads into $\mathbf{A}$, including incident bidirected edges, and all outgoing directed edges from $\mathbf{B}$. For disjoint sets of vertices $\mathbf{A}_1, \cdots, \mathbf{A}_m \subseteq \mathbf{A}$, we use $(\mathbf{A}_1, \cdots, \mathbf{A}_m)$ to represent an ordered set, where the order needs to be highlighted. For the unordered set, we use braces as $\{A_1, \cdots, A_m\}$.

Adjustment Criterion. We revisit the problem of characterizing a criterion such that the causal effect is given as an adjustment. Formally, the *adjustment set* relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ is defined as a set $\mathbf{Z}$ such that, for any P compatible with $\mathcal{G}$, the following is satisfied (Shpitser et al., 2010):

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}). \tag{1}$$

The *adjustment criterion* (Shpitser et al., 2010) provides a sound and complete criterion for the causal effect $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to be expressed as a covariate adjustment in Eq. (1). To present the adjustment criterion, we define a *proper causal path* between $\mathbf{X}$ and $\mathbf{Y}$ as a directed path from any of $X \in \mathbf{X}$ to any of $Y \in \mathbf{Y}$ on which the path does not contain a node in $\mathbf{X} \setminus \{X\}$. A proper causal path set $\mathrm{pcp}_{\mathcal{G}}(\mathbf{X}, \mathbf{Y})$ is a collection of nodes excluding $\mathbf{X}$ on all proper causal paths from $X \in \mathbf{X}$ to $Y \in \mathbf{Y}$, defined in Table 2. $\mathrm{dpcp}_{\mathcal{G}}(\mathbf{X}, \mathbf{Y})$ is a descendant of the proper causal path set defined in Table 2. The adjustment criterion utilizes this subgraph called the *proper back-door graph*:

Definition 1 (Proper Back-Door Graph (van der Zander et al., 2014))**.** *Let $(\mathbf{X}, \mathbf{Y})$ denote a disjoint pair in $\mathbf{V}$. The proper back-door graph $\mathcal{G}_{\mathrm{pbd}}^{\mathbf{X},\mathbf{Y}}$ is a graph obtained from $\mathcal{G}$ by removing an edge $X \rightarrow D$ for each $X \in \mathbf{X}$ and $D \in \mathrm{pcp}_{\mathcal{G}}(\mathbf{X}, \mathbf{Y})$.*

Then, the adjustment criterion and its characterization are given as follows.

Definition 2 (Adjustment Criterion (Shpitser et al., 2010; van der Zander et al., 2014))**.** *Let* $(\mathbf{X}, \mathbf{Y})$ *denote a disjoint pair in* $\mathbf{V}$*. A set of variables* $\mathbf{Z} \subseteq \mathbf{V} \setminus (\mathbf{X} \cup \mathbf{Y})$ *is said to satisfy the adjustment criterion relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$ *if (1)* $(\mathbf{Y} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{pbd}}}$ *and (2)* $\mathbf{Z} \cap \text{dpcp}_{\mathcal{G}}(\mathbf{X}, \mathbf{Y}) = \emptyset$*.*

Proposition 1 (Shpitser et al. (2010); van der Zander et al. (2014))**.** $\mathbf{Z}$ *satisfies the adjustment criterion relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$ *if and only if* $\mathbf{Z}$ *is an adjustment set relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$*.*

For concreteness, in Fig. 3, $\mathbf{Z} \coloneqq \{Z_a, Z_b\}$ satisfies the adjustment criterion relative to $\mathbf{X} \coloneqq \{X_1, X_2\}$ and $\mathbf{Y} \coloneqq \{Y_1, Y_2\}$. See (van der Zander et al., 2014, Sec. 4.2) for details. In the later sections, we will extend this result to the sequential context.

3 Limitations of Existing Criterion for Sequential Covariate Adjustment

In this section, we demonstrate the limitations of existing criteria for sequential covariate adjustment, which is defined as follows:

Definition 3 (Sequential Covariate Adjustment)**.** *Let* $(\mathbf{X}, \mathbf{Y})$ *denote a pair of ordered sets such that* $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_m)$ *and* $\mathbf{Y} = (\mathbf{Y}_0, \ldots, \mathbf{Y}_m)$*. Let* $\mathbf{Z} \subseteq \mathbf{V} \setminus (\mathbf{X} \cup \mathbf{Y})$ *denote vertices ordered as* $\mathbf{Z} = (\mathbf{Z}_1, \ldots, \mathbf{Z}_m)$*. Define* $\mathbf{H}_i \coloneqq \mathbf{X}^{(i)} \cup \mathbf{Y}^{(i)} \cup \mathbf{Z}^{(i)}$*. The set* $\mathbf{Z}$ *is said to be a sequential adjustment set relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$ *if for any* P *compatible with* $\mathcal{G}$*, the following is satisfied*[3]*:*

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} \prod_{j=0}^{m} P(\mathbf{z}_{j+1}, \mathbf{y}_j \mid \mathbf{h}_{j-1}, \mathbf{x}_j, \mathbf{z}_j). \tag{2}$$

The right-hand side of Eq. (2) *is called the sequential covariate adjustment of* $\mathbf{Z}$ *w.r.t.* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$*.*

In this definition, sequential covariate adjustment relies on specific partitions $\mathbf{X} = (\mathbf{X}_1, \cdots, \mathbf{X}_m)$ and $\mathbf{Y} = (\mathbf{Y}_0, \cdots, \mathbf{Y}_m)$. To standardize these partitions, we employ a fixed structure based on a topological order of $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ so as to ensure that the partition aligns with the causal structure.

Definition 4 (Partitioning Operator)**.** *Let* $\mathbf{X}$ *and* $\mathbf{Y}$ *denote disjoint vertices where* $\mathbf{X} = (X_1, \cdots, X_m)$ *is topologically ordered along with* $\mathcal{G}$*. Then, the operator* $\text{PT}^{\mathbf{X}}_{\mathcal{G}}$ *partitions* $\mathbf{Y}$ *relative to* $\mathbf{X}$ *and* $\mathcal{G}$*; i.e.,* $\text{PT}^{\mathbf{X}}_{\mathcal{G}}(\mathbf{Y}) = (\mathbf{Y}_0, \cdots, \mathbf{Y}_m)$*, where*

$$\mathbf{Y}_i \coloneqq \begin{cases} \mathbf{Y} \setminus \text{De}_{\mathcal{G}}(\mathbf{X}) & \text{for } i = 0, \\ \big((\mathbf{Y} \setminus \mathbf{Y}^{(i-1)}) \cap \text{De}_{\mathcal{G}}(X_i)\big) \setminus \text{De}_{\mathcal{G}}(\mathbf{X}^{\geq i+1}) & \text{for } i = 1, \cdots, m. \end{cases} \tag{3}$$

Hereafter, $\mathbf{X} = (X_1, \cdots, X_m)$ consists of distinct scalar treatment vertices in a fixed topological order of $\mathcal{G}$, and the outcome blocks are $\text{PT}^{\mathbf{X}}_{\mathcal{G}}(\mathbf{Y})$. The results concern pairwise disjoint covariate blocks outside $\mathbf{X} \cup \mathbf{Y}$ with $\mathbf{Z}_i \cap \text{De}_{\mathcal{G}}(\mathbf{X}^{\geq i+1}) = \emptyset$, called admissible blocks in Definition 7. A block may contain descendants of its current treatment. Every vertex of $\mathbf{H}_{i-1}$ is then a non-descendant of X_i and all later treatments.

By the chain rule, the supplied expression in Eq. (2) is equivalently

$$\sum_{\mathbf{z}} P(\mathbf{y}_0) \prod_{i=1}^{m} P(\mathbf{z}_i \mid \mathbf{h}_{i-1}) P(\mathbf{y}_i \mid \mathbf{h}_{i-1}, x_i, \mathbf{z}_i).$$

For each fixed treatment value it is a normalized distribution: generate the blocks in the order $\mathbf{Y}_0, \mathbf{Z}_1, \mathbf{Y}_1, \ldots, \mathbf{Z}_m, \mathbf{Y}_m$, inserting x_i immediately before generating $\mathbf{Y}_i$. Universal validity asks whether this distribution equals the joint-intervention distribution for every compatible positive causal model.

3.1 Limitation of mSBD Criterion

The multi-outcome sequential back-door (mSBD) criterion (Jung et al., 2020) has been developed to determine whether an ordered set $\mathbf{Z} = (\mathbf{Z}_1, \cdots, \mathbf{Z}_m)$ constitutes a sequential adjustment set:

[3] If an ordered set associates with an index out of range (e.g., $\mathbf{X}_0$, $\mathbf{X}_{m+1}$), it is interpreted as the empty set. An interval $[a, b]$ (e.g., those associated with $\sum$ or $\bigcup$) with $b < a$ is also an empty set.

Definition 5 (Multi-outcome Sequential Back-Door Criterion (Jung et al., 2020)). *Let* $(\mathbf{X}, \mathbf{Y})$ *denote a pair of ordered sets in* $\mathcal{G}$. *Let* $\mathbf{Z} \subseteq \mathbf{V} \setminus (\mathbf{X} \cup \mathbf{Y})$ *be ordered as* $\mathbf{Z} = (\mathbf{Z}_1, \dots, \mathbf{Z}_m)$, *and let* $\mathbf{H}_i := \mathbf{X}^{(i)} \cup \mathbf{Y}^{(i)} \cup \mathbf{Z}^{(i)}$. *Then,* $\mathbf{Z}$ *is said to satisfy the multi-outcome sequential back-door (mSBD) criterion w.r.t.* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$ *if, for* $i = 1, \dots, m$,

$$\mathbf{Z}_i \text{ is non-descendant of } \mathbf{X}^{\geq i} \text{ in } \mathcal{G}; \text{ and} \tag{4}$$

$$\left(\mathbf{Y}^{\geq i} \perp\!\!\!\perp X_i \mid \mathbf{H}_{i-1}, \mathbf{Z}_i\right)_{\mathcal{G}_{\underline{X_i}\overline{\mathbf{X}^{\geq i+1}}}}. \tag{5}$$

Proposition 2 (mSBD adjustment (Jung et al., 2020)). *If* $\mathbf{Z} = (\mathbf{Z}_1, \dots, \mathbf{Z}_m)$ *satisfies the mSBD criterion w.r.t.* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$, *the causal effect* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ *is identifiable and expressible as sequential covariate adjustment of* $\mathbf{Z}$ *w.r.t.* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$:

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} \prod_{j=0}^{m} P(\mathbf{z}_{j+1}, \mathbf{y}_j \mid \mathbf{h}_{j-1}, x_j, \mathbf{z}_j). \tag{6}$$

For concreteness, consider the causal diagram $\mathcal{G}$ in Fig. 1 with $\mathbf{X} = (X_1, X_2)$ and $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$. Applying the partitioning operator $\mathrm{PT}_{\mathcal{G}}^{\mathbf{X}}$, we divide $\mathbf{Y}$ into two subsets: $\mathbf{Y}_1 = \{Y_1\}$ and $\mathbf{Y}_2 = \{Y_2\}$. Next, we define $\mathbf{Z} := (\mathbf{Z}_1, \mathbf{Z}_2)$, where $\mathbf{Z}_1 := \{Z_a, Z_b\}$ and $\mathbf{Z}_2 := \{Z_c, Z_d\}$. This ordered set $\mathbf{Z}$ satisfies the mSBD criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$. The first condition of the mSBD criterion holds since $\mathbf{Z}_1$ does not include any descendants of X_1 in $\mathcal{G}$, and similarly, $\mathbf{Z}_2$ excludes any descendants of X_2. To check the second condition of the mSBD criterion, consider the graph $\mathcal{G}_{\underline{X_1}\overline{X_2}}$ derived from $\mathcal{G}$ by removing the edges $X_1 \to Y_1$ and $\{Z_b \to X_2, Z_c \to X_2, Y_1 \to X_2\}$. In this graph, $(\mathbf{Y}_1, \mathbf{Y}_2)$ are conditionally independent of X_1 given $\mathbf{Z}_1$, because any path from X_1 to $\{Y_1, Y_2\}$ is blocked (d-separated) by $\mathbf{Z}_1$. Finally, in the graph $\mathcal{G}_{\underline{X_2}}$, where the edge $X_2 \to Y_2$ is removed, any paths between X_2 and Y_2 are d-separated when conditioned on $\mathbf{H}_1 \cup \mathbf{Z}_2$, where $\mathbf{H}_1 = \{X_1, Y_1\} \cup \mathbf{Z}_1$. This analysis confirms that $\mathbf{Z}$ satisfies the mSBD criterion, and therefore, the causal effect is given as follows:

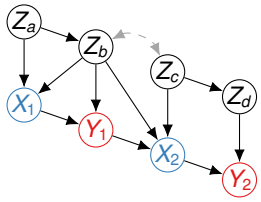


Figure 1: Graph exemplifying mSBD in Def. 5

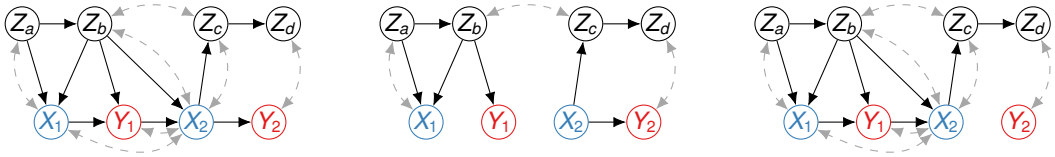
$$\begin{aligned} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) &= \sum_{\mathbf{z}_1, \mathbf{z}_2} P(\mathbf{y}_2 \mid \mathbf{x}_1, \mathbf{x}_2, \mathbf{y}_1, \mathbf{z}_1, \mathbf{z}_2) P(\mathbf{z}_2, \mathbf{y}_1 \mid \mathbf{x}_1, \mathbf{z}_1) P(\mathbf{z}_1) \\ &= \sum_{z_a, z_b, z_c, z_d} P(y_2 \mid x_1, x_2, y_1, z_a, z_b, z_c, z_d) P(z_c, z_d, y_1 \mid x_1, z_a, z_b) P(z_a, z_b). \end{aligned} \tag{7}$$

Despite its soundness (Prop. 2), the mSBD criterion is not complete. In other words, there are causal graphs such that $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is expressible through sequential covariate adjustment, even though $\mathbf{Z} := (\mathbf{Z}_1, \dots, \mathbf{Z}_m)$ does not satisfy the mSBD criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$.

Proposition 3 (Incompleteness of mSBD criterion). *There exists a causal graph* $\mathcal{G}$ *where the causal effect* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ *can be represented as a sequential covariate adjustment with* $\mathbf{Z}$ *such that* $\mathbf{Z}$ *does not satisfy the mSBD criterion relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$.

To illustrate the incompleteness of the mSBD criterion, consider the graph $\mathcal{G}$ in Fig. 2a with $\mathbf{X} = (X_1, X_2)$ and $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$ (i.e., $\mathbf{Y}_0 := \emptyset$) with $\mathbf{Y}_1 = \{Y_1\}$ and $\mathbf{Y}_2 = \{Y_2\}$. Set $\mathbf{Z} := (\mathbf{Z}_1, \mathbf{Z}_2)$, where $\mathbf{Z}_1 := \{Z_a, Z_b\}$ and $\mathbf{Z}_2 := \{Z_c, Z_d\}$. Then, while $\mathbf{Z}_1$ satisfies both Eqs. (4, 5) of the mSBD criterion, $\mathbf{Z}_2$ does not satisfy the mSBD condition in Eq. (4) since $\mathbf{Z}_2$ includes the descendant of X_2. Nevertheless, the causal effect $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is still expressible through sequential covariate adjustment as in Eq. (7). To demonstrate, we begin with the observation that $\mathbf{Z}_1$ satisfies the mSBD criterion in Eqs. (4, 5), leading to $P(y_1, y_2 \mid \mathrm{do}(x_1, x_2)) = \sum_{z_a, z_b} P(y_2 \mid \mathrm{do}(x_2), \mathbf{h}_1) P(y_1 \mid x_1, z_a, z_b) P(z_a, z_b)$, with $\mathbf{H}_1 := \mathbf{X}_1 \cup \mathbf{Y}_1 \cup \mathbf{Z}_1 = \{X_1, Y_1, Z_a, Z_b\}$. Next,

$$\begin{aligned} P(y_2 \mid \mathrm{do}(x_2), \mathbf{h}_1) &= \sum_{z_c} P(y_2 \mid \mathrm{do}(x_2), \mathbf{h}_1) P(z_c \mid \mathbf{h}_1) \\ &= \sum_{z_c} P(y_2 \mid \mathrm{do}(x_2), \mathbf{h}_1, z_c) P(z_c \mid \mathbf{h}_1), \end{aligned}$$



(a) SAC (b) $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$ (c) $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$

Figure 2: (a) Graph exemplifying SAC in Def. 7 and (b,c) its proper sequential back-door graphs

where the last equality holds by Rule 1 of do-calculus (Pearl, 1995): $(Y_2 \perp\!\!\!\perp Z_c \mid X_2, \mathbf{H}_1)_{\mathcal{G}_{\overline{X_2}}}$. Further,

$$P(y_2 \mid \text{do}(x_2), \mathbf{h}_1, z_c) = \sum_{z_d} P(y_2 \mid \text{do}(x_2), \mathbf{h}_1, z_c, z_d) P(z_d \mid \text{do}(x_2), \mathbf{h}_1, z_c)$$
$$= \sum_{z_d} P(y_2 \mid x_2, \mathbf{h}_1, z_c, z_d) P(z_d \mid \mathbf{h}_1, z_c),$$

where the last equality holds by Rule 2 and 3 of do-calculus, i.e., $(Y_2 \perp\!\!\!\perp X_2 \mid \mathbf{H}_1, Z_c, Z_d)_{\mathcal{G}_{\underline{X_2}}}$ and $(Z_d \perp\!\!\!\perp X_2 \mid \mathbf{H}_1, Z_c)_{\mathcal{G}}$, respectively. In Rule 3, X_2 is an ancestor of the conditioned variable Z_c, so its incoming edges are retained for this check. Combining these derivations, $P(y_1, y_2 \mid \text{do}(x_1, x_2))$ is expressible as the same as in Eq. (7). This exemplifies the incompleteness of mSBD criterion.

3.2 Limitation of Other Existing Criteria

Besides mSBD, there are other methods attempting to establish conditions for determining if the causal effect can be represented as sequential covariate adjustment. For instance, the *no-unmeasured-confounding assumption* (Rosenbaum and Rubin, 1983) is a sufficient verification tool. However, as illustrated in Fig. 2a, the causal effect is expressible through sequential covariate adjustment even though an unmeasured confounder between the treatment X_2 and the outcome Y_1 presents.

On the other hand, the *sequential ignorability* criterion (Robins, 1986, 2000) provides a condition for $\mathbf{Z}$ to be a sequential adjustment set:

$$\mathbf{Y}^{\geq i}(\mathbf{x}) \perp\!\!\!\perp X_i \mid \mathbf{Z}_i, \mathbf{H}_{i-1}, \tag{8}$$

where $\mathbf{Y}^{\geq i}(\mathbf{x})$ is a counterfactual variable induced from $\mathbf{Y}^{\geq i}$ by fixing $\mathbf{X} = \mathbf{x}$ in the SCM. Graphical methods such as *single world intervention graphs* (SWIGs) (Richardson and Robins, 2013) and *twin-networks* (Pearl, 2000; Shpitser and Pearl, 2008) help in analyzing such counterfactual variables. These representations require care when factual and counterfactual variables are considered together. SWIGs, for example, face challenges in capturing $\mathbf{Z}_1, \mathbf{Z}_2$ as a sequential adjustment set in Fig 2a, because they fail to accommodate both $Y_2(x_2)$ and $\mathbf{Z}_2$ in a SWIG. This limitation arises since the counterfactual variable $\mathbf{Z}_2(x_2)$, instead of the observable $\mathbf{Z}_2$, will appear on the SWIG. In contrast, twin-networks are capable of exploring the relationship between $Y_2(x_2)$ and $\mathbf{Z}_2$. A factual world and one fixed joint-intervention world require two submodel graphs; the number of treatment coordinates alone does not imply an exponentially large twin network (Shpitser and Pearl, 2008, Sec. 4). Additional distinct intervention worlds may require additional submodels. These observations motivate a direct graphical characterization for the adjustment formula considered here.

4 Sequential Adjustment Criterion

In this section, we give the corrected *sequential adjustment criterion* (SAC), which characterizes universal validity of the admissible sequential adjustment expression in Eq. (2). To begin with, we define a special subgraph called *proper sequential back-door graph*:

Definition 6 (Proper Sequential Back-Door Graph)**.** *For a disjoint pair* $(\mathbf{X}, \mathbf{Y})$ *and the index* i*, the proper sequential back-door graph* $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},i}$ *is a proper back-door graph induced from* $\mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}$ *relative to* $(X_i, \mathbf{Y}^{\geq i})$*; i.e.,*

$$\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},i} := (\mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}})^{X_i, \mathbf{Y}^{\geq i}}_{\text{pbd}}. \tag{9}$$

As constructing the proper back-door graph from $\mathcal{G}$ takes linear time $O(|\mathbf{V}| + |\mathbf{E}|)$ (van der Zander et al., 2014), the sequential back-door graph can also be constructed in linear time $O(|\mathbf{V}| + |\mathbf{E}|)$. Equipped with $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},i}$, the *sequential adjustment criterion* is presented as follows:

Definition 7 (Sequential Adjustment Criterion (SAC))**.** *Let $(\mathbf{X}, \mathbf{Y})$ denote a disjoint pair. Let $\mathbf{Z} \coloneqq (\mathbf{Z}_1, \cdots, \mathbf{Z}_m)$ consist of pairwise disjoint blocks disjoint from $\mathbf{X} \cup \mathbf{Y}$, where each $\mathbf{Z}_i$ contains no descendant of $\mathbf{X}^{\geq i+1}$. We call these blocks admissible. Their stage assignment need not be a topological order of the covariates; the treatment order and this nondescendancy condition are the restrictions used below. Then, $\mathbf{Z}$ is said to satisfy sequential adjustment criterion (SAC) w.r.t. $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if the following conditions are satisfied for $i = 1, \cdots, m$:*

$$(\mathbf{Y}^{\geq i} \perp\!\!\!\perp X_i \mid \mathbf{Z}_i, \mathbf{H}_{i-1})_{\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},i}}; \text{ and} \tag{10}$$

$$\mathbf{Z}_i \cap \text{dpcp}_{\mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}}(X_i, \mathbf{Y}^{\geq i}) = \emptyset. \tag{11}$$

Equation (11) corrects the graph used for the published forbidden set. Both conditions are evaluated after removing incoming arrowheads into later treatments; Eq. (10) already used this graph in the published version. The future-descendant restriction defining admissibility remains in the original graph $\mathcal{G}$. If $\mathbf{Y}^{\geq i} = \emptyset$, both stage conditions are vacuous. Hereafter SAC denotes the corrected criterion.

The corrected criterion is sound and complete for admissible sequential covariate adjustment:

Theorem 1 (Soundness and Completeness)**.** *Let $(\mathbf{X}, \mathbf{Y})$ denote a disjoint pair with the treatment order and outcome partition specified above. An admissible ordered sequence $\mathbf{Z}$ satisfies SAC relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if and only if it is a sequential adjustment set; that is, for every compatible positive causal model,*

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} \prod_{j=0}^{m} P(\mathbf{z}_{j+1}, \mathbf{y}_j \mid \mathbf{h}_{j-1}, x_j, \mathbf{z}_j). \tag{12}$$

If SAC fails, a compatible positive linear Gaussian model and treatment values make the right-hand side incorrect at the mean of one outcome coordinate.

Why the published forbidden set must change. Consider the DAG $X_1 \to Z \to X_2 \to Y$, with $\mathbf{Y}_0 = \mathbf{Y}_1 = \emptyset$, $\mathbf{Y}_2 = \{Y\}$, and $(\mathbf{Z}_1, \mathbf{Z}_2) = (\{Z\}, \emptyset)$. These blocks are admissible. Every compatible positive model satisfies

$$\sum_{z} P(z)P(y \mid x_1, x_2, z) = P(y \mid x_2) = P(y \mid \text{do}(x_1, x_2)).$$

The published condition rejects this sequence because $Z \in \text{dpcp}_{\mathcal{G}}(X_1, Y)$. Its path through the later treatment X_2 is removed under the joint intervention. In contrast, $\text{dpcp}_{\mathcal{G}_{\overline{X_2}}}(X_1, Y) = \emptyset$, and the sequence satisfies corrected SAC. The example refutes necessity of the published condition. Necessity of the replacement requires an additional argument: Lemma S.2 constructs a countermodel for every failed stage and proves that marginalizing the earlier history cannot remove its error.

Gaussian countermodels. Theorem 1 implies that a fixed admissible formula is universally valid over the positive nonparametric model class if and only if it is valid in every compatible positive linear Gaussian model. This follows from the Gaussian countermodel in the necessity proof. Agreement in one or finitely many Gaussian models does not establish this universal statement.

For instance, consider Fig. 2a, where we witnessed that the causal effect $P(y_1, y_2 \mid \text{do}(x_1, x_2))$ is given as a sequential covariate adjustment of $\mathbf{Z} = (\mathbf{Z}_1 = \{Z_a, Z_b\}, \mathbf{Z}_2 = \{Z_c, Z_d\})$. To confirm that this $\mathbf{Z}$ satisfies SAC relative to $(\mathbf{X}, \mathbf{Y})$ for $\mathbf{X} = (X_1, X_2)$ and $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$ with $\mathbf{Y}_1 = \{Y_1\}$ and $\mathbf{Y}_2 = \{Y_2\}$, we examine $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$ (Fig. 2b) and $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$ (Fig. 2c). In $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$, we observe that $\mathbf{Z}_1 = \{Z_a, Z_b\}$ d-separates X_1 and $\{Y_1, Y_2\}$. Moreover, $\mathbf{Z}_1$ does not contain any descendant of proper causal paths from X_1 to $\{Y_1, Y_2\}$ in $\mathcal{G}_{\overline{X_2}}$. Similarly, in $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$, the path between X_2 and Y_2 is d-separated conditioned on $\mathbf{Z}_2 = \{Z_c, Z_d\}$ and $\mathbf{H}_1 = \{Z_a, Z_b, X_1, Y_1\}$. Additionally, $\mathbf{Z}_2$ is not on any descendant of proper causal paths between X_2 and Y_2. Consequently, $\mathbf{Z}$ fulfills SAC relative to $(\mathbf{X}, \mathbf{Y})$ in the graph $\mathcal{G}$.

SAC includes the mSBD criterion and admits a partition of every ordinary adjustment set. The following proofs also give direct graphical arguments for these inclusions.

Corollary 1 (mSBD $\implies$ SAC)**.** *If* $\mathbf{Z}$ *satisfies the mSBD criterion relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$*, then* $\mathbf{Z}$ *satisfies the SAC relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$*.*

To demonstrate, consider $\mathcal{G}$ in Fig. 1, where $\mathbf{Z} \coloneqq (\mathbf{Z}_1, \mathbf{Z}_2)$ with $\mathbf{Z}_1 \coloneqq \{Z_a, Z_b\}$ and $\mathbf{Z}_2 \coloneqq \{Z_c, Z_d\}$. This $\mathbf{Z}$ satisfies the mSBD criterion relative to $(\mathbf{X}, \mathbf{Y})$ for $\mathbf{X} = (X_1, X_2)$ and $\mathbf{Y} = (Y_1, Y_2)$. To ensure that $\mathbf{Z}$ also satisfies the SAC relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$, we verify that $(\mathbf{Z}_1, \mathbf{Z}_2)$ meet the conditions in Eqs. (10, 11), respectively. Since $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are both non-descendants of X_1 and X_2, respectively, we only need to verify that they meet the conditions of Eq. (10) for $(X_1, (Y_1, Y_2))$ and (X_2, Y_2) respectively. First, consider $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$, which removes edges $\{X_1 \to Y_1, Z_b \to X_2, Z_c \to X_2, Y_1 \to X_2\}$ from $\mathcal{G}$. In this graph, X_1 is d-separated from (Y_1, Y_2) given $\mathbf{Z}_1$. Next, consider $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$, which removes the edge $\{X_2 \to Y_2\}$ from $\mathcal{G}$. In this graph, X_2 is d-separated from Y_2 given $(\mathbf{Z}_1, X_1, Y_1)$ and $\mathbf{Z}_2$. Therefore, $\mathbf{Z}$ also satisfies the SAC relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$.

The broader coverage of SAC compared to the mSBD criterion is evident when compared to AC. First, the mSBD criterion is not extensive enough to cover the AC criterion. To witness, consider Fig. 3. Here, $\mathbf{Z} \coloneqq \{Z_a, Z_b\}$ satisfies AC relative to $(\mathbf{X}, \mathbf{Y})$ for $\mathbf{X} = (X_1, X_2)$ and $\mathbf{Y} = (Y_1, Y_2)$. However, $\mathbf{Z}$ does not satisfy the mSBD criterion since it is a descendant set of $\mathbf{X}_1 \coloneqq \{X_1\}$. In contrast, AC is a sufficient criterion for SAC.

Theorem 2 (AC $\implies$ SAC)**.** *If* $\mathbf{Z}$ *satisfies the AC relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$*, then there exists a partition of* $\mathbf{Z}$ *and a topological order of* $\mathbf{X}$ *such that* $\mathbf{Z}$ *satisfies the SAC relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$*.*

For concreteness, consider $\mathcal{G}$ in Fig. 3 (van der Zander et al., 2014). In $\mathcal{G}$, $\mathbf{Z} \coloneqq \{Z_a, Z_b\}$ satisfies the adjustment criterion relative to $(\mathbf{X}, \mathbf{Y})$ for $\mathbf{X} = (X_1, X_2)$ and $\mathbf{Y} = (Y_1, Y_2)$ in $\mathcal{G}$, and the causal effect is given as an adjustment as follows:

$$P(y_1, y_2 \mid \mathrm{do}(x_1, x_2)) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}). \tag{13}$$

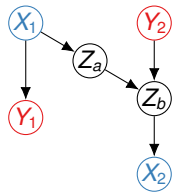


Figure 3: (van der Zander et al., 2014, Fig. 2)

Thm. 2 implies that $\mathbf{Z}$ meets the SAC relative to $(\mathbf{X}, \mathbf{Y})$ in graph $\mathcal{G}$. To demonstrate this, for $\mathbf{X} = (X_1, X_2)$, we apply the partitioning operator in Def. 4 and obtain $\mathbf{Y}_0 = \{Y_2\}$, $\mathbf{Y}_1 \coloneqq \{Y_1\}$, and $\mathbf{Y}_2 = \emptyset$. We set $\mathbf{Z}_1 \coloneqq \{Z_a, Z_b\}$. Since $\mathbf{Y}_2$ is empty, it is sufficient to demonstrate that $\mathbf{Z}_1$ meets the conditions of Eqs. (10) and (11). First, $\mathbf{Z}_1$ meets the condition of Eq. (11) as it contains no descendants on the causal paths from X_1 to $\mathbf{Y}_1$. Consider the modified graph $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$, obtained by removing the edges $\{X_1 \to Y_1, Z_b \to X_2\}$. In this reduced graph, X_1 and $\mathbf{Y}_1$ are d-separated given $\mathbf{Z}_1$ and $\mathbf{Y}_0$, confirming the conditions of SAC. Therefore, $\mathbf{Z} \coloneqq (\mathbf{Z}_1, \mathbf{Z}_2)$ satisfies SAC relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$, and the causal effect is described as

$$P(y_1, y_2 \mid \mathrm{do}(x_1, x_2)) = \sum_{z_a, z_b} P(y_1 \mid x_1, z_a, z_b, y_2) P(z_a, z_b \mid y_2) P(y_2). \tag{14}$$

Eqs. (13, 14) are equivalent since $P(y_1 \mid x_1, x_2, z_a, z_b, y_2) = P(y_1 \mid x_1, z_a, z_b, y_2)$ and $P(y_2 \mid x_1, x_2, z_a, z_b) = P(y_2 \mid z_a, z_b)$ by the conditional independence implied by $\mathcal{G}$.

4.1 Constructive Sequential Adjustment Criterion

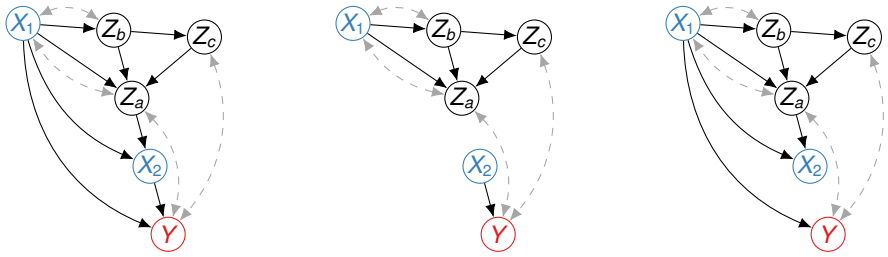
The sequential adjustment criterion in Definition 7 checks a supplied admissible partition $\mathbf{Z} = (\mathbf{Z}_1, \ldots, \mathbf{Z}_m)$. The construction below finds a valid admissible sequence whenever one exists. Its graphical existence equivalence becomes an equivalence for universal formula validity by Theorem 1.

Theorem 3 (Construction of Sequential Adjustment Set)**.** *Let* $(\mathbf{X}, \mathbf{Y})$ *denote a disjoint pair in* $\mathcal{G}$*. Define* $\mathbf{Z}^{\text{an}} \coloneqq (\mathbf{Z}^{\text{an}}_1, \cdots, \mathbf{Z}^{\text{an}}_m)$ *and corresponding* $(\mathbf{F}^{\text{an}}_1, \cdots, \mathbf{F}^{\text{an}}_m)$ *and* $(\mathbf{H}^{\text{an}}_0, \cdots, \mathbf{H}^{\text{an}}_{m-1})$ *alternatively as follows: For* $i = 1, \cdots, m$*,*

$$\mathbf{H}^{\text{an}}_{i-1} \coloneqq \mathbf{X}^{(i-1)} \cup \mathbf{Y}^{(i-1)} \cup \bigcup_{j=1}^{i-1} \mathbf{Z}^{\text{an}}_j, \tag{15}$$

$$\mathbf{F}^{\text{an}}_i \coloneqq \mathbf{X} \cup \mathbf{Y} \cup \mathbf{H}^{\text{an}}_{i-1} \cup \mathrm{dpcp}_{\mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}}(X_i, \mathbf{Y}^{\geq i}) \cup \mathrm{De}_{\mathcal{G}}(\mathbf{X}^{\geq i+1}), \tag{16}$$

$$\mathbf{Z}^{\text{an}}_i \coloneqq \mathrm{An}_{\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},i}}(\{X_i\} \cup \mathbf{Y}^{\geq i} \cup \mathbf{H}^{\text{an}}_{i-1}) \setminus \mathbf{F}^{\text{an}}_i. \tag{17}$$



(a) $\mathcal{G}$ (b) $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$ (c) $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$

Figure 4: Example for `minSCA` and corresponding proper sequential back-door graphs

Then, the following statements are equivalent:

1. *There exists an admissible sequential adjustment set* $\mathbf{Z} = (\mathbf{Z}_1, \ldots, \mathbf{Z}_m)$ *relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$.
2. *The ordered set* $\mathbf{Z}^{\text{an}}$ *satisfies SAC relative to* $(\mathbf{X}, \mathbf{Y})$ *in* $\mathcal{G}$.

The set $\mathbf{Z}^{\text{an}}$ can be constructed in $O(m(|\mathbf{V}| + |\mathbf{E}|))$ since the procedures for constructing $\text{dpcp}_{\mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}}(X_i, \mathbf{Y}^{\geq i})$ and $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},i}$ in Eq. (16), as well as finding ancestors and descendants in Eqs. (16, 17), each require $O(|\mathbf{V}| + |\mathbf{E}|)$ and are repeated m times.

For the graph in Fig. 2a, the canonical construction gives $\mathbf{Z}^{\text{an}}_1 = \{Z_a, Z_b\}$ and $\mathbf{Z}^{\text{an}}_2 = \emptyset$. In particular, Z_c and Z_d are descendants of the future treatment X_2 and are excluded from the first block by Eq. (16); they are not directed ancestors of the first-stage endpoints in Fig. 2b either. The resulting sequence satisfies SAC: the first block separates X_1 from $\{Y_1, Y_2\}$, and at the second stage $\{X_1, Y_1, Z_a, Z_b\}$ already separates X_2 from Y_2 in Fig. 2c. The published example incorrectly included Z_c, Z_d in the first canonical block.

As an example that the second statement implies the first one, consider Fig. 4a, where $\mathbf{Y}_0 := \emptyset$, $\mathbf{Y}_1 := \emptyset$ and $\mathbf{Y}_2 := \{Y\}$ by the mechanism of the partitioning operator $\text{PT}^{\mathbf{X}}_{\mathcal{G}}$. Next, $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$ and $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$ are depicted in Figs. (4b, 4c). Then, $\mathbf{Z}^{\text{an}}_1 := \emptyset$, since none of $\{Z_a, Z_b, Z_c\}$ are the ancestors of $\{X_1, Y\}$. $\mathbf{Z}^{\text{an}}_1$ satisfies Eqs. (10, 11) in the SAC, since the X_1 and Y are d-separated in $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},1}$ given $\mathbf{Z}^{\text{an}}_1$. Next, $\mathbf{Z}^{\text{an}}_2 := \{Z_a, Z_b, Z_c\}$ since they are ancestors of X_2 in $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$. This $\mathbf{Z}^{\text{an}}_2$ satisfies Eqs. (10, 11), since X_2 and Y are d-separated in $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\text{psbd},2}$ given $\mathbf{Z}^{\text{an}}_2 \cup \{X_1\}$. Therefore, $\mathbf{Z}^{\text{an}} := (\mathbf{Z}^{\text{an}}_1, \mathbf{Z}^{\text{an}}_2) = (\emptyset, \{Z_a, Z_b, Z_c\})$ satisfies SAC relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$.

4.2 Minimal Sequential Adjustment Criterion

We have demonstrated the effectiveness of constructing sequential adjustment sets. However, when dealing with large graphs, $\mathbf{Z}^{\text{an}}$ may include a large number of vertices, leading to high computational costs when evaluating sequential covariate adjustments. This situation highlights the need for a more parsimonious adjustment set. To address this, we introduce the *minimal* sequential covariate adjustment set, which is inclusion-minimal at each stage conditional on its retained prefix. This is different from minimum cardinality or minimum estimation variance.

Definition 8 (Minimal Sequential Covariate Adjustment Set)**.** *Let* $(\mathbf{X}, \mathbf{Y})$ *denote a disjoint pair in* $\mathcal{G}$. *An ordered set of vertices* $\mathbf{Z}^{\min} := (\mathbf{Z}^{\min}_1, \cdots, \mathbf{Z}^{\min}_m)$ *with admissible blocks is said to be a minimal sequential covariate adjustment set if, for each* $i = 1, \cdots, m$,

1. $\mathbf{Z}^{\min}_i$ *satisfies Eqs. (10, 11).*
2. *For any* $\mathbf{Z}'_i \subsetneq \mathbf{Z}^{\min}_i$, $\mathbf{Z}'_i$ *does not satisfy Eqs. (10, 11) with the earlier retained history held fixed.*

The published algorithm used two closure operations at each stage. The closure must account for a mandatory history $\mathbf{H}$.

Algorithm 1: $\texttt{minSCA}(\mathbf{X}, \mathbf{Y}, \mathcal{G})$ with finalized histories

Input: A causal graph $\mathcal{G}$, topologically ordered treatments $\mathbf{X} = (X_1, \ldots, X_m)$, and outcomes partitioned by $\mathrm{PT}_{\mathcal{G}}^{\mathbf{X}}$

Output: A minimal sequential adjustment set, or $\perp$ if none exists

$\mathbf{H} \coloneqq \mathbf{Y}_0$
for $i = 1, \ldots, m$ **do**
  $\mathcal{G}_i \coloneqq \mathcal{G}_{\mathrm{psbd},i}^{\mathbf{X},\mathbf{Y}}$
  $\mathbf{F}_i \coloneqq \mathbf{X} \cup \mathbf{Y} \cup \mathbf{H} \cup \mathrm{dpcp}_{\mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}}(X_i, \mathbf{Y}^{\geq i}) \cup \mathrm{De}_{\mathcal{G}}(\mathbf{X}^{\geq i+1})$
  $\mathbf{A}_i \coloneqq \mathrm{An}_{\mathcal{G}_i}(\{X_i\} \cup \mathbf{Y}^{\geq i} \cup \mathbf{H}) \setminus \mathbf{F}_i$
  if $\mathbf{Y}^{\geq i} = \emptyset$ **then**
    $\mathbf{Z}_i^{\min} \coloneqq \emptyset$
  else
    Form a DAG $\widetilde{\mathcal{G}}_i$ by replacing each bidirected edge in $\mathcal{G}_i$ with a distinct latent parent of its endpoints
    $\mathbf{S}_i \coloneqq \mathrm{FindMinSep}(\widetilde{\mathcal{G}}_i, \{X_i\}, \mathbf{Y}^{\geq i}, \mathbf{I} = \mathbf{H}, \mathbf{R} = \mathbf{H} \cup \mathbf{A}_i)$
    if $\mathbf{S}_i = \perp$ **then**
      return $\perp$
    end
    $\mathbf{Z}_i^{\min} \coloneqq \mathbf{S}_i \setminus \mathbf{H}$
  end
  $\mathbf{H} \coloneqq \mathbf{H} \cup \{X_i\} \cup \mathbf{Y}_i \cup \mathbf{Z}_i^{\min}$
end
return $(\mathbf{Z}_1^{\min}, \ldots, \mathbf{Z}_m^{\min})$

Definition 9 (Closure with a Required Conditioning Set (van der Zander and Liśkiewicz, 2020))**.** *Let* $\mathbf{A}, \mathbf{B}, \mathbf{C}$ *be pairwise disjoint, and let* $\mathbf{H} \subseteq \mathbf{C}$*. Put* $\mathbf{L} = \mathrm{An}_{\mathcal{G}}(\mathbf{A} \cup \mathbf{B} \cup \mathbf{H})$*. Define* $\mathrm{closure}_{\mathcal{G}}^{\mathbf{H}}(\mathbf{A}; \mathbf{B}, \mathbf{C})$ *as* $\mathbf{A}$ *together with every vertex reachable from* $\mathbf{A}$ *by a path in* $\mathcal{G}[\mathbf{L}]$ *whose internal non-colliders lie outside* $\mathbf{C}$*. Path endpoints are exempt from that restriction. No additional collider-activation condition is imposed.*

This closure is not the set of vertices d-connected to $\mathbf{A}$ given $\mathbf{C}$. For example, in $X \to K \leftarrow Y$, $K \to B$, the empty-conditioning closure of X relative to B contains Y, although X and Y are d-separated. The required history determines the ancestral domain and must remain included when the candidate conditioning set changes. For DAGs this is the closure used in the constrained separator method of van der Zander and Liśkiewicz (2020); directed–bidirected graphs can be handled by latent-parent expansion, with latent vertices excluded from eligible adjustment sets.

Two failures of the published pruning algorithm. First consider $X_1 \to X_2$, $C \to X_2$, $C \to Y$, $X_2 \to Y$. The canonical sequence is $(\{C\}, \emptyset)$. The original algorithm removes C from the first block, but evaluates the second block using the unpruned history containing C. It returns $(\emptyset, \emptyset)$, leaving $X_2 \leftarrow C \to Y$ open. A valid stagewise minimal sequence is $(\emptyset, \{C\})$.

A separate failure already occurs with one treatment. Consider $A \to X$, $A \to C \leftarrow B$, $B \to Y$, and $X \to Y$, with outcomes $\{C, Y\}$. Here $\mathbf{Y}_0 = \{C\}$ and $\mathbf{Y}_1 = \{Y\}$. The original closure domain $\mathrm{An}_{\mathcal{G}_{\mathrm{pbd}}}(\{X, Y\})$ omits the mandatory history vertex C. The two pruning intersections return the empty set, although $X \leftarrow A \to C \leftarrow B \to Y$ is active given C. Including the history in the ancestral domain repairs this local defect, but does not repair the simultaneous-history defect in the preceding example.

Here $\mathrm{FindMinSep}$ is the constrained separator procedure of van der Zander and Liśkiewicz (2020, Proposition 5.2): it returns a separator $\mathbf{S}$ satisfying $\mathbf{I} \subseteq \mathbf{S} \subseteq \mathbf{R}$ that is inclusion-minimal subject to retaining $\mathbf{I}$, or returns $\perp$ when none exists. The latent vertices in $\widetilde{\mathcal{G}}_i$ do not belong to $\mathbf{R}$. Its ancestor search includes the required history: $\mathrm{An}_{\widetilde{\mathcal{G}}_i}(\{X_i\} \cup \mathbf{Y}^{\geq i} \cup \mathbf{H})$. The history used to construct the next candidate set contains only the covariates actually retained in the returned earlier blocks. The stage graph and the descendants of proper causal paths are computed after intervening on future treatments, using the fixed treatment order and outcome partition; only the history and the associated permitted candidate set change as blocks are finalized.

Theorem 4 (Construction of Minimal Sequential Adjustment Sets)**.** *There exists an admissible sequential adjustment set if and only if Algorithm 1 returns a sequence. Every returned sequence satisfies Definition 8. No sequence obtained by componentwise deletion of its covariates, with at least one strict deletion, is a sequential adjustment set. The running time is* $O(m(|\mathbf{V}| + |\mathbf{E}|))$*. Admissible sequences have pairwise disjoint blocks, are disjoint from* $\mathbf{X} \cup \mathbf{Y}$*, and satisfy* $\mathbf{Z}_i \cap \mathrm{De}_{\mathcal{G}}(\mathbf{X}^{\geq i+1}) = \emptyset$ *for every* i*.*

The minimality here is inclusion-minimality for each block with its earlier history fixed. It does not imply minimum total cardinality, minimum variance, or a componentwise subset of the unpruned canonical sequence. The existence and componentwise minimality claims concern all universally valid formulas within the stated admissible class, by Theorem 1.

For Fig. 4a, the initial ancestor construction gives $(\emptyset, \{Z_a, Z_b, Z_c\})$, and the repaired forward algorithm returns $(\emptyset, \{Z_a\})$. The four-node counterexample above shows why the next candidate set must be rebuilt: after removing C from the first block, the algorithm selects it at the second stage.

5 Conclusion

The corrected SAC is sound and complete for the stated admissible sequential adjustment formulas in a known causal DAG or ADMG. Its forbidden descendants must be computed after intervening on later treatments. A replacement necessity proof isolates future treatments and lifts a Gaussian adjustment countermodel through the sequential prefix, establishing failure of the final marginalized formula. The corrected arguments retain the inclusion of mSBD and ordinary adjustment. The canonical construction and repaired forward algorithm find a valid sequence whenever one exists; finalizing each history and retaining mandatory conditioning variables gives stagewise and componentwise inclusion-minimality in linear time per treatment. These corrections restore the original completeness and construction guarantees within their explicit scope. Prescribed finite-domain necessity, nonadmissible covariate allocations, other outcome schedules, uncertain graphs, and variance-optimal sequences require separate results.

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Supplement to "Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference"

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A Proofs

The following arguments replace the corresponding proofs in the published version.

A.1 Proof of Theorem 1

We use the causal graph and positivity conventions of Sec. 2. Lemmas S.1 and S.2 establish the two implications for the same fixed graph, treatment order, outcome partition, and admissible sequence.

Conditional adjustment for a causal latent-projection graph. Conditional adjustment given non-descendants is studied by LaPlante and Perković (2024). The soundness identity needed here also follows from Shpitser et al. (2010, Theorem 4): if $\mathbf{W} \cup \mathbf{Z}$ satisfies ordinary adjustment for a single treatment X and outcomes $\mathbf{T}$, then $\mathbf{T}(x) \perp\!\!\!\perp X \mid \mathbf{W}, \mathbf{Z}$. When $\mathbf{W}$ consists of non-descendants of X, the tower property and consistency give

$$P(\mathbf{t} \mid \mathrm{do}(x), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{T}(x) = \mathbf{t} \mid \mathbf{w}, \mathbf{z}) P(\mathbf{z} \mid \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{t} \mid x, \mathbf{w}, \mathbf{z}) P(\mathbf{z} \mid \mathbf{w}).$$

The cited SCM result applies to causal latent-projection graphs. Deterministic latent mechanisms are allowed; only the observed conditionals used here require positivity. For a single treatment, its forbidden-set convention agrees with the descendants of non-treatment vertices on directed paths used here. Thus the soundness argument does not restrict the compatible models to the particular latent-parent expansion used below for countermodels.

Lemma S.1 (Soundness of SAC)**.** *Let $\mathbf{X} = (X_1, \ldots, X_m)$ be topologically ordered, let $\mathbf{Y} = \mathrm{PT}_{\mathcal{G}}^{\mathbf{X}}(\mathbf{Y})$, and let the pairwise disjoint sets $\mathbf{Z}_1, \ldots, \mathbf{Z}_m$ be disjoint from $(\mathbf{X}, \mathbf{Y})$, with $\mathbf{Z}_i \cap \mathrm{De}_{\mathcal{G}}(\mathbf{X}^{\geq i+1}) = \emptyset$. If $\mathbf{Z}$ satisfies SAC, then it is a sequential adjustment set.*

Proof. Fix i, and put $\mathcal{G}_i = \mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}$ and $\mathbf{H} = \mathbf{H}_{i-1}$. Every vertex in $\mathbf{H}$ is a non-descendant of $\mathbf{X}^{\geq i}$: this follows respectively from the treatment order, the outcome partition, and the restriction on earlier covariate sets. Corrected SAC supplies the forbidden-node restriction and proper-back-door separation in the same graph $\mathcal{G}_i$.

To apply conditional adjustment without treating fixed intervention nodes as ordinary positive random variables, replace the mechanisms of the future treatments $\mathbf{X}^{\geq i+1}$ by mutually independent randomizations with positive full support, independent of all other exogenous variables. Denote the resulting distribution by Q_i. Its graph is $\mathcal{G}_i$; conditioning the randomized treatments on $\mathbf{x}^{\geq i+1}$ gives the same distribution of the other variables as intervening on those treatments in the original model. In $\mathcal{G}_i$, future treatment vertices have no incoming arrowheads. Adding them to a separating set cannot activate a collider, since they have no proper ancestors. Consequently the SAC separation also holds after conditioning on their values. They are non-descendants of X_i in $\mathcal{G}_i$.

Both $\mathbf{H}$ and the future-treatment roots are non-descendants of X_i. Therefore $\mathbf{H} \cup \mathbf{X}^{\geq i+1} \cup \mathbf{Z}_i$ satisfies ordinary adjustment for $(X_i, \mathbf{Y}^{\geq i})$ in $\mathcal{G}_i$. Apply the preceding conditional-adjustment identity in Q_i with $\mathbf{W} = \mathbf{H} \cup \mathbf{X}^{\geq i+1}$, and condition the randomized future treatments on $\mathbf{x}^{\geq i+1}$. This gives

$$P(\mathbf{y}^{\geq i} \mid \mathrm{do}(\mathbf{x}^{\geq i}), \mathbf{h}) = \sum_{\mathbf{z}_i} P(\mathbf{y}^{\geq i} \mid \mathrm{do}(\mathbf{x}^{\geq i+1}), x_i, \mathbf{z}_i, \mathbf{h}) P(\mathbf{z}_i \mid \mathbf{h}). \tag{A.1}$$

Here $Q_i(\mathbf{z}_i \mid \mathbf{h}, \mathbf{x}^{\geq i+1}) = P(\mathbf{z}_i \mid \mathbf{h})$, because $\mathbf{Z}_i \cup \mathbf{H}$ consists of non-descendants of all future treatments in the original graph. By the same argument, the joint distribution of $\mathbf{Y}_i \cup \{X_i\} \cup \mathbf{Z}_i \cup \mathbf{H}$ is unaffected by those future interventions. Splitting the outcome factor in the preceding display therefore yields

$$\begin{aligned} &P(\mathbf{y}^{\geq i} \mid \mathrm{do}(\mathbf{x}^{\geq i}), \mathbf{h}_{i-1}) \\ &\quad = \sum_{\mathbf{z}_i} P(\mathbf{y}^{\geq i+1} \mid \mathrm{do}(\mathbf{x}^{\geq i+1}), \mathbf{h}_i) P(\mathbf{y}_i \mid \mathbf{h}_{i-1}, x_i, \mathbf{z}_i) P(\mathbf{z}_i \mid \mathbf{h}_{i-1}). \end{aligned} \tag{A.2}$$

Finally $P(\mathbf{y}_0 \mid \mathrm{do}(\mathbf{x})) = P(\mathbf{y}_0)$. Iterating the displayed identity from $i = 1$ to m gives

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y}_0) \prod_{i=1}^{m} P(\mathbf{z}_i \mid \mathbf{h}_{i-1}) P(\mathbf{y}_i \mid \mathbf{h}_{i-1}, x_i, \mathbf{z}_i)$$
$$= \sum_{\mathbf{z}} \prod_{j=0}^{m} P(\mathbf{z}_{j+1}, \mathbf{y}_j \mid \mathbf{h}_{j-1}, x_j, \mathbf{z}_j),$$

where empty endpoint sets have the conventions of Definition 3. This proves soundness. □

Lemma S.2 (Completeness of SAC)**.** *Fix a known causal DAG or ADMG $\mathcal{G}$, the specified treatment order and outcome partition, and an admissible sequence $\mathbf{Z}$. If $\mathbf{Z}$ fails corrected SAC in $\mathcal{G}$, there is a compatible positive linear Gaussian model in which Eq.* (2) *fails, already at the mean of one outcome coordinate. In particular, $\mathbf{Z}$ is not a sequential adjustment set in this same graph $\mathcal{G}$.*

Proof. Fix a failing stage i, and write

$$\mathcal{G}_i = \mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}, \qquad \mathbf{C} = \mathbf{H}_{i-1} \cup \mathbf{Z}_i, \qquad \mathbf{T} = \mathbf{Y}^{\geq i}.$$

Failure of a conditional identity at this stage would not alone prove failure after marginalizing earlier covariates. The following three claims provide a countermodel for the final expression.

Claim 1: the failure survives isolation of future treatments. Let $\mathcal{G}_i^0$ be obtained from $\mathcal{G}_i$ by also removing outgoing edges of all future treatments. These treatments are then isolated. We show that $\mathbf{C}$ fails ordinary adjustment for $(X_i, \mathbf{T})$ in $\mathcal{G}_i^0$.

First consider a DAG. Both X_i and $\mathbf{C}$ contain no descendant of a future treatment in the original graph. A directed path beginning at X_i cannot enter a future-treatment root in $\mathcal{G}_i$. The causal paths defining its forbidden set and proper back-door deletion are therefore unchanged by isolation. Directed continuations from these paths to forbidden covariates cannot enter such roots either.

Now suppose a path from X_i to $\mathbf{T}$ is active given $\mathbf{C}$ in the proper back-door graph of $\mathcal{G}_i$. If it passed through a future-treatment root R, follow the path from R toward X_i. Either it remains directed all the way, contradicting the treatment order, or its first collider is a descendant of R. Activating that collider would require a descendant of R in $\mathbf{C}$, contrary to admissibility and the history restrictions. The active path therefore avoids future treatments. Its directed collider-activation paths avoid them as well, since no directed path can enter these roots. The separation failure survives isolation.

Finally, $\mathbf{H}_{i-1}$ consists of non-descendants of X_i, so it cannot contribute a forbidden vertex. A failure of Eq. (11) makes $\mathbf{C}$ fail the ordinary forbidden-set condition. Otherwise, the surviving separation failure makes it fail ordinary adjustment by Definition 2. For an ADMG, use its latent-parent DAG expansion; the observed directed ancestry and separation statements are preserved.

Claim 2: ordinary-adjustment failure has a Gaussian mean witness. If $\mathbf{C}$ fails ordinary adjustment for a single treatment X and outcome set $\mathbf{T}$ in a causal DAG, there is a centered nondegenerate linear Gaussian structural equation model and $Y^* \in \mathbf{T}$ such that

$$\beta_{Y^*, X \mid \mathbf{C}} \neq \tau_{X \to Y^*},$$

where β is the coefficient of X in the regression on $(X, \mathbf{C})$ and τ is the total causal effect. This is the stronger witness established in the proof of Perkovic et al. (2018, Appendix E, Theorem 57, pp. 44–51). That proof constructs a centered positive Gaussian model whose ordinary adjusted mean differs from the interventional mean at treatment value one. The former is $\beta_{Y^*, X \mid \mathbf{C}}$ because $\mathbb{E}[\mathbf{C}] = 0$, and the latter is $\tau_{X \to Y^*}$. The result allows descendants of X in $\mathbf{C}$ and covers both forbidden-set and blocking failures. Latent vertices may be included outside the adjustment and outcome sets.

Apply this result to $\mathcal{G}_i^0$, or its latent-parent DAG expansion. The resulting observed distribution is positive, and future treatments are independent of all other variables. The model is compatible with

the original $\mathcal{G}$: deleted edges have coefficient zero. Throughout the argument, keep the outcome and covariate blocks specified in the original graph; their assignment is not recomputed in this submodel.

Claim 3: averaging the prefix preserves the incorrect slope. Write $\Phi_{\mathbf{Z}}(P; \mathbf{x}, \mathbf{y})$ for the right-hand side of Eq. (2). By its equivalent chain-rule form in Sec. 3,

$$\Phi_{\mathbf{Z}}(P; \mathbf{x}, \mathbf{y}) = \sum_{\mathbf{z}} P(\mathbf{y}_0) \prod_{k=1}^{m} P(\mathbf{z}_k \mid \mathbf{h}_{k-1}) P(\mathbf{y}_k \mid \mathbf{h}_{k-1}, x_k, \mathbf{z}_k).$$

In the witness model, the isolated future treatments drop out of all observational conditionals. The kernels from the $\mathbf{Y}_i$ factor onward form a chain-rule factorization. Summing the later covariates reduces that suffix to

$$P(\mathbf{y}^{\geq i} \mid x_i, \mathbf{h}_{i-1}, \mathbf{z}_i).$$

The preceding kernels, including $P(\mathbf{z}_i \mid \mathbf{h}_{i-1})$, generate a normalized distribution $Q_{\mathbf{x}^{(i-1)}}$ on $\mathbf{L} = \mathbf{Y}^{(i-1)} \cup \mathbf{Z}^{(i)}$. Past treatments are fixed, and this distribution does not depend on x_i. It has finite moments because it is generated by finitely many affine Gaussian conditional kernels.

For the outcome coordinate in Claim 2, the observational conditional mean is

$$\mathbb{E}[Y^* \mid x_i, \mathbf{c}] = \beta x_i + \gamma^\top \mathbf{c}, \qquad \beta = \beta_{Y^*, X_i \mid \mathbf{C}}.$$

Marginalizing every other outcome in $\Phi_{\mathbf{Z}}$ therefore gives mean

$$\beta x_i + \int \gamma^\top \mathbf{c} \, Q_{\mathbf{x}^{(i-1)}}(d\boldsymbol{\ell}),$$

where $\mathbf{c}$ includes the fixed earlier treatment values. Its slope in x_i is β, regardless of whether any earlier stage satisfies SAC. Setting the earlier treatment values to zero makes the prefix centered by induction, so the integral is then zero.

The true joint-intervention mean has slope $\tau_{X_i \to Y^*}$. No directed path from X_i to Y^* can pass through an earlier treatment, by topological order, and future treatments are isolated. Intervening on the other treatments therefore removes none of the paths contributing to this total effect. Set $x_i = 1$ and every other treatment to zero. The sequential mean is β and the interventional mean is τ, which differ by Claim 2. This proves failure of the final formula in the fixed original graph. If compatibility requires every graph edge to have a nonzero coefficient, sufficiently small perturbations of the deleted coefficients preserve the strict mean discrepancy by continuity. No correctness assumption on the earlier stages was used. □

Together, Lemmas S.1 and S.2 prove Theorem 1. They also prove the stated equivalence between validity in all compatible positive Gaussian models and universal nonparametric validity for this admissible formula class.

A.2 Proof of Corollary 1

Suppose $\mathbf{Z}$ satisfies mSBD. At each stage, $\mathbf{C}_i = \mathbf{H}_{i-1} \cup \mathbf{Z}_i$ contains no descendants of X_i. In particular, Eq. (11) holds. Let $\mathcal{G}_i = \mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}$. If an active path from X_i to $\mathbf{Y}^{\geq i}$ given $\mathbf{C}_i$ existed in the proper back-door graph of $\mathcal{G}_i$, it could not be directed from X_i to its outcome endpoint. If its first edge pointed out of X_i, the first collider on the path would be a descendant of X_i with a descendant in $\mathbf{C}_i$, contradicting the non-descendant restriction. Thus its first edge does not point out of X_i. Its collider-activation paths to $\mathbf{C}_i$ cannot use an outgoing edge of X_i either, for the same reason. The path would therefore remain active in $\mathcal{G}_{\underline{X_i}\overline{\mathbf{X}^{\geq i+1}}}$, contradicting mSBD. This proves Eq. (10), and hence the corollary, without invoking a completeness claim.

A.3 Proof of Theorem 2

Assume $\mathbf{Z}$ satisfies ordinary AC relative to the unordered treatment set $\mathbf{X}$ and outcome set $\mathbf{Y}$. Choose any topological order $\mathbf{X} = (X_1, \ldots, X_m)$, and partition the outcomes as in Definition 4. For each

$Z \in \mathbf{Z}$, define

$$t(Z) = \max\big(\{k : Z \in \mathrm{De}_{\mathcal{G}}(X_k)\} \cup \{0\}\big), \qquad \mathbf{Z}_i = \{Z \in \mathbf{Z} : \min(t(Z)+1, m) = i\}.$$

These sets partition $\mathbf{Z}$ and are non-descendants of their respective future treatments. They also respect the directed order of covariates: if one covariate is an ancestor of another, its value of t cannot be larger. For $i < m$, $\mathbf{Z}_i$ contains no descendants of X_i, so Eq. (11) holds. At $i = m$, every causal path from X_m to an outcome is proper relative to the entire treatment set, because it cannot contain another treatment. Since $\mathbf{Y}_m$ contains every outcome descended from X_m, ordinary AC gives Eq. (11) at the last stage as well.

It remains to prove Eq. (10). Write

$$\mathcal{G}_i = \mathcal{G}_{\overline{\mathbf{X}^{\geq i+1}}}, \qquad \mathbf{C}_i = \mathbf{H}_{i-1} \cup \mathbf{Z}_i, \qquad \mathcal{B} = \mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\mathrm{pbd}}.$$

Suppose, to obtain a contradiction, that a path π from X_i to some $Y \in \mathbf{Y}^{\geq i}$ is active given $\mathbf{C}_i$ in the proper back-door graph of $\mathcal{G}_i$.

First, π contains no future treatment. Indeed, future treatments have only outgoing edges in $\mathcal{G}_i$. Starting at such an internal vertex and following π toward X_i would either produce a directed path to X_i, contradicting the treatment order, or encounter a collider descended from that future treatment. The collider could not be activated, since $\mathbf{C}_i$ has no descendants of future treatments. Any earlier treatment on π must instead be a collider, because earlier treatments belong to $\mathbf{C}_i$.

Next, every edge of π is retained in $\mathcal{B}$. The edges deleted in forming $\mathcal{B}$ have a treatment as their source. An earlier treatment on π has no outgoing edge on π, and future treatments do not occur on π. An edge out of X_i that is deleted in forming $\mathcal{B}$ begins a proper causal path to an outcome. That path avoids all other treatments, ends in $\mathbf{Y}^{\geq i}$, and is retained in $\mathcal{G}_i$ before forming its proper back-door graph. Its first edge was therefore already removed there and cannot belong to π.

Collider activation is retained in $\mathcal{B}$ as well. For each collider of π, take a directed path to its first vertex in $\mathbf{C}_i$. A removed edge on that path cannot leave an earlier treatment, since that treatment already belongs to $\mathbf{C}_i$, and cannot leave a future treatment, since no future treatment is an ancestor of $\mathbf{C}_i$. If $i < m$, it cannot leave X_i either, because $\mathbf{C}_i$ contains no descendants of X_i. If $i = m$, an edge out of X_m removed in forming $\mathcal{B}$ cannot lead to $\mathbf{H}_{m-1}$, which consists of non-descendants of X_m, and cannot lead to $\mathbf{Z}_m$, since that would make the latter contain a descendant of a global proper causal path, contrary to ordinary AC. Hence π is active given $\mathbf{C}_i$ in $\mathcal{B}$.

Finally, adding the later covariates $\mathbf{Z}^{\geq i+1}$ cannot block π. There is nothing to prove when $i = m$. Otherwise, suppose a later covariate Z occurs as a non-collider on π. By the chosen partition, Z is a descendant of some treatment X_k with $k \geq i$. A non-collider on an active path is an ancestor of an endpoint or of a conditioning vertex: follow a tail along the path until reaching an endpoint or an activated collider. It cannot be an ancestor of X_i, by acyclicity and the treatment order, and cannot be an ancestor of $\mathbf{C}_i$, since $\mathbf{C}_i$ contains no descendants of X_k. Therefore there is a directed path from Z to the outcome endpoint in the stage graph. This path contains no treatment: later treatments have no incoming edges, and the treatment order excludes earlier treatments and X_i. Append this path to a directed path from X_k to Z, retaining only the segment beginning at its last treatment. The resulting path is a proper causal path from the full treatment set to $\mathbf{Y}$ and contains Z. This contradicts ordinary AC. No later covariate can consequently occur as a non-collider on π, so π remains active given $\mathbf{Z} \cup \mathbf{H}_{i-1}$ in $\mathcal{B}$.

But ordinary AC states $(\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z})_{\mathcal{B}}$. Decomposition and weak union imply

$$(X_i \perp\!\!\!\perp \mathbf{Y}^{\geq i} \mid \mathbf{Z}, \mathbf{X}^{(i-1)}, \mathbf{Y}^{(i-1)})_{\mathcal{B}},$$

contradicting the active path just obtained. This proves Eq. (10) at every stage and completes the constructive proof.

A.4 Proof of Theorem 3

Lemma A.1 (Constrained Ancestor Separator (van der Zander et al., 2014, Lemma 3.4))**.** *Let* $\mathbf{A}, \mathbf{B}, \mathbf{I}, \mathbf{R}$ *satisfy* $\mathbf{I} \subseteq \mathbf{R}$ *and* $\mathbf{R} \cap (\mathbf{A} \cup \mathbf{B}) = \emptyset$*. If some d-separator* $\mathbf{C}$ *satisfies* $\mathbf{I} \subseteq \mathbf{C} \subseteq \mathbf{R}$*, then* $\mathrm{An}_{\mathcal{G}}(\mathbf{A} \cup \mathbf{B} \cup \mathbf{I}) \cap \mathbf{R}$ *also separates* $\mathbf{A}$ *and* $\mathbf{B}$*.*

For a directed–bidirected graph, apply the DAG version to a latent-parent expansion, keeping latent vertices outside $\mathbf{R}$. To see the DAG argument, put $\mathbf{T} = \mathrm{An}_{\mathcal{G}}(\mathbf{A} \cup \mathbf{B} \cup \mathbf{I})$. Restricting a valid separator to $\mathbf{T}$ preserves separation in the smaller ancestral moral graph. Adding the other allowed vertices in $\mathbf{T}$ then preserves separation because the ancestor set is fixed and extra conditioning only deletes vertices.

Lemma A.2 (Construction for a Fixed History)**.** *Fix an admissible prefix* $(\mathbf{Z}_1, \ldots, \mathbf{Z}_{i-1})$ *and its history* $\mathbf{H}_{i-1}$*. The canonical block in Eq.* (17)*, evaluated with this history, satisfies the two stage-*i *SAC conditions if and only if some allowed stage-*i *block does. This is an equivalence between graphical conditions, not a claim of necessity for a conditional adjustment formula.*

Proof. Only the reverse implication needs proof. Apply Lemma A.1 in $\mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\mathrm{psbd},i}$ with endpoints $\{X_i\}, \mathbf{Y}^{\geq i}$, required set $\mathbf{I} = \mathbf{H}_{i-1}$, and allowed set $\mathbf{R} = \mathbf{H}_{i-1} \cup (\mathbf{V} \setminus \mathbf{F}_i)$, where $\mathbf{F}_i$ is Eq. (16) with the fixed history. The assumed separator is $\mathbf{H}_{i-1} \cup \mathbf{Z}_i$. The lemma returns $\mathbf{H}_{i-1} \cup \mathbf{Z}^{\mathrm{an}}_i$. The remaining restrictions hold by the subtraction of $\mathbf{F}_i$. □

Proof of Theorem 3. By Theorem 1, the first statement is equivalent to existence of an admissible sequence satisfying SAC. The reverse implication follows from the same theorem. For the forward implication, let $\mathbf{Z} = (\mathbf{Z}_1, \ldots, \mathbf{Z}_m)$ satisfy SAC. Write $\mathcal{G}_i = \mathcal{G}^{\mathbf{X},\mathbf{Y}}_{\mathrm{psbd},i}$ and fix a stage i.

First, every covariate in an earlier canonical block $\mathbf{Z}^{\mathrm{an}}_j$, $j < i$, is an ancestor in $\mathcal{G}_i$ of $\mathbf{X}^{(i-1)} \cup \mathbf{Y}$. To see this, recursively substitute the definition of the earlier canonical histories in the ancestor construction. At stage j, the resulting directed paths end in $\mathbf{X}^{(j)} \cup \mathbf{Y}$. Such paths cannot pass through a future treatment: all incoming edges to those treatments were removed in $\mathcal{G}_j$, and their starting covariate is not a treatment. Passing from $\mathcal{G}_j$ to $\mathcal{G}_i$ restores some edges and removes only outgoing edges of X_i that were not already removed. The directed paths in question do not contain X_i, so they remain in $\mathcal{G}_i$. This also justifies the recursive substitution by induction over j.

Let $\mathbf{C} = \mathbf{Z}_i \cup \mathbf{H}_{i-1}$ be the separator furnished by the given sequence. The preceding observation implies

$$\mathbf{H}^{\mathrm{an}}_{i-1} \setminus \mathbf{C} \subseteq \mathrm{An}_{\mathcal{G}_i}(\{X_i\} \cup \mathbf{Y}^{\geq i} \cup \mathbf{C}).$$

Adding ancestors of the endpoints and the conditioning set to a separator preserves separation. Indeed, in the ancestral moral graph the ancestor set does not change, and the additional conditioning only removes more vertices. For directed–bidirected graphs, the same argument applies after representing each bidirected edge by a separate latent parent. Consequently $\mathbf{C} \cup \mathbf{H}^{\mathrm{an}}_{i-1}$ separates X_i and $\mathbf{Y}^{\geq i}$ in $\mathcal{G}_i$.

Define

$$\mathbf{Z}'_i = \left(\mathbf{Z}_i \cup \bigcup_{j<i} \mathbf{Z}_j \right) \setminus \mathbf{H}^{\mathrm{an}}_{i-1}.$$

Then $\mathbf{Z}'_i \cup \mathbf{H}^{\mathrm{an}}_{i-1} = \mathbf{C} \cup \mathbf{H}^{\mathrm{an}}_{i-1}$. The vertices contributed by earlier blocks are non-descendants of X_i and all later treatments. Thus they avoid the forbidden set at stage i; the original $\mathbf{Z}_i$ already avoids that set. Therefore $\mathbf{Z}'_i$ is a permitted separator supplement for the canonical history. The constrained ancestor-separator lemma now shows that the canonical $\mathbf{Z}^{\mathrm{an}}_i$ also separates the stage endpoints given that history. Its other admissibility conditions hold by construction. This holds for every i, proving that the canonical sequence satisfies SAC whenever any admissible sequence satisfies SAC. □

A.5 Proof of Theorem 4

Proof. By Theorem 1, universal validity and corrected SAC are equivalent for every admissible sequence. We first prove that finalizing a stage can preserve a valid completion. Suppose a fixed

prefix through stage $i-1$ has a completion $(\mathbf{Z}_i, \ldots, \mathbf{Z}_m)$ satisfying SAC. Let $\mathbf{H}$ be its fixed history, and let $\mathbf{A}_i$ be the candidate set in the algorithm. The constrained ancestor-separator lemma ensures that $\mathbf{H} \cup \mathbf{A}_i$ separates the stage endpoints whenever a permitted supplement for $\mathbf{H}$ exists. Thus the local separator procedure can return a set $\mathbf{M} \subseteq \mathbf{A}_i$ for which $\mathbf{H} \cup \mathbf{M}$ separates the endpoints. The same assertion holds with $\mathbf{M} = \emptyset$ when the outcome set is empty.

For $i < m$, replace the completion by

$$\mathbf{Z}'_i = \mathbf{M}, \qquad \mathbf{Z}'_{i+1} = (\mathbf{Z}_{i+1} \setminus \mathbf{M}) \cup (\mathbf{Z}_i \setminus \mathbf{M}), \qquad \mathbf{Z}'_k = \mathbf{Z}_k \setminus \mathbf{M} \quad (k > i+1).$$

All earlier blocks stay fixed. This is an admissible sequence: the new block $\mathbf{M}$ satisfies the restrictions at stage i, and the only vertices transferred to a later block come from $\mathbf{Z}_i$. They are non-descendants of X_{i+1} and every later treatment, so they avoid both the forbidden set and the future-descendant restriction at stage $i+1$. All other modified blocks are subsets of their original blocks.

For $k > i$, let $\mathbf{C}_k = \mathbf{H}_{k-1} \cup \mathbf{Z}_k$ denote the old conditioning set. The modified conditioning set is exactly

$$\mathbf{C}'_k = \mathbf{C}_k \cup \mathbf{M}.$$

By its construction, every vertex of $\mathbf{M}$ is an ancestor in $\mathcal{G}_i$ of $\{X_i\} \cup \mathbf{Y}^{\geq i} \cup \mathbf{H}$. The corresponding directed path contains no future treatment, because $\mathcal{G}_i$ has no incoming edges to those treatments and its starting vertex is not a treatment. On passing from $\mathcal{G}_i$ to $\mathcal{G}_k$, previously removed edges may be restored, and new deletions affect only outgoing edges of X_k. The directed path therefore survives. Its endpoint belongs to $\{X_k\} \cup \mathbf{Y}^{\geq k} \cup \mathbf{C}_k$, since the earlier treatments, earlier outcomes, and fixed prefix are contained in $\mathbf{C}_k$. Consequently

$$\mathbf{M} \subseteq \mathrm{An}_{\mathcal{G}_k}(\{X_k\} \cup \mathbf{Y}^{\geq k} \cup \mathbf{C}_k).$$

Adding such ancestors to a separating set preserves separation: the ancestral moral graph does not change, and additional conditioning only removes vertices. For a directed–bidirected graph, this argument applies to its latent-variable DAG representation. Hence $\mathbf{C}'_k$ separates the endpoints at every later stage. The modified sequence remains SAC-valid. For $i = m$ there is no later stage, and the new valid local block itself completes the sequence.

Starting from any SAC-valid sequence, the preceding argument proves inductively that every prefix finalized by the algorithm still admits a SAC-valid completion. The algorithm therefore cannot fail when a valid sequence exists. Conversely, each returned block satisfies the local SAC restrictions with its actual finalized history, so a returned sequence satisfies SAC. Local constrained minimality means that no proper subset of that block remains a separator with the same history. The forbidden-node restriction is preserved by taking subsets; thus the returned sequence satisfies Definition 8.

For any strict componentwise deletion, take the earliest stage whose block changes. Its earlier history is unchanged, and its smaller block fails the local minimality condition. The modified sequence therefore fails SAC and, by Theorem 1, is not universally valid. This proves the claimed componentwise inclusion-minimality.

Finally, each stage graph, forbidden set, and ancestor set can be constructed in linear time. Replacing a bidirected edge by one latent root adds one vertex and two edges, so the expanded DAG has $O(|\mathbf{V}| + |\mathbf{E}|)$ size. The cited constrained separator procedure is linear in this size. There are m stages, giving the stated bound. □

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Changes

2026-10-25. Yonghan Jung, Min Woo Park and Sanghack Lee, *Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference*, Advances in Neural Information Processing Systems 37 (2024). Original publication and proceedings record.

This unofficial correction was prepared with AI assistance. **Author review pending.** It is not a publisher-issued correction or an approval by the coauthors.

Location	Before → After	Assumptions and limits
Definition 7, Eq. (11); Theorem 1	Forbidden descendants in $\mathcal{G} \rightarrow$ compute both causal paths and descendants in $\mathcal{G}_{\overline{\mathbf{X}_{\geq i+1}}}$. Eq. (10) already used this intervention. The published necessity claim fails on $X_1 \rightarrow Z \rightarrow X_2 \rightarrow Y$ with $(\{Z\}, \emptyset)$; the corrected criterion accepts it.	Distinct scalar treatments in fixed topological order; graph-derived outcome blocks. Covariate blocks are disjoint, outside treatments/outcomes, and exclude descendants of later treatments in original $\mathcal{G}$. The theorem explicitly uses Definition 7's timing restriction; interblock topological order is not required.
Theorem 1; Lemmas S.1–S.2; Sec. 2	Path-composition and local-necessity arguments → conditional adjustment after randomizing future treatments, and a Gaussian countermodel whose erroneous treatment slope survives prefix marginalization.	Known causal DAG/latent-projection ADMG, including bows, with positivity where formula conditionals are evaluated. Universal validity includes real-valued Gaussian models; necessity for prescribed finite alphabets is not proved. No latent-density positivity assumption.
Corollary 1; Theorems 2–3; Lemmas A.1–A.2	Unsupported changes of quantifiers and conditioning sets → explicit allocation of ordinary adjustment sets, constrained ancestor separation, and proofs that added history preserves separation.	Existence and inclusion results concern the same admissible formula class. Ancestors of required history are included in the search domain.
Algorithm 1; Theorem 4; Definitions 8–9	Pruning blocks against a stale history → finalize each block before recomputing the next. Use a constrained separator with required history; check feasibility. Closure reachability is no longer equated with d-connection.	Stagewise and componentwise inclusion minimality, not minimum cardinality or variance. $O(m(\lvert\mathbf{V}\rvert + \lvert\mathbf{E}\rvert))$ assumes the cited linear-time constrained separator routine.
Sec. 4.1, Fig. 2a example; Secs. 3–4	First canonical block $\{Z_a, Z_b, Z_c, Z_d\} \rightarrow \{Z_a, Z_b\}$, second block empty. Correct stage conditioning and the Rule 3 graph. Remove the unsupported exponential twin-network size claim.	Z_c, Z_d descend from future X_2. A fixed factual/intervention pair does not require exponentially many worlds.
Throughout	Correct algebraic indices, equation references and notation; supply supporting proof citations.	Main theorem, definition and equation numbers are preserved.

Experimental scope. The original paper contains graphical examples, not an empirical benchmark. The affected construction example is corrected in place. No experiments or downstream empirical studies were rerun for this edition. Earlier finite counterexample and implementation checks do not establish universal validity or downstream empirical performance.

Citation. Cite the original 2024 paper. If this version is used, add "Unofficial AI-corrected version dated 2026-10-25 consulted" to the reference note, and the permanent URL: `https://www.sanghacklee.me/ai-corrected/SAC_AI_corrected_2026-10-25.pdf`.

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